

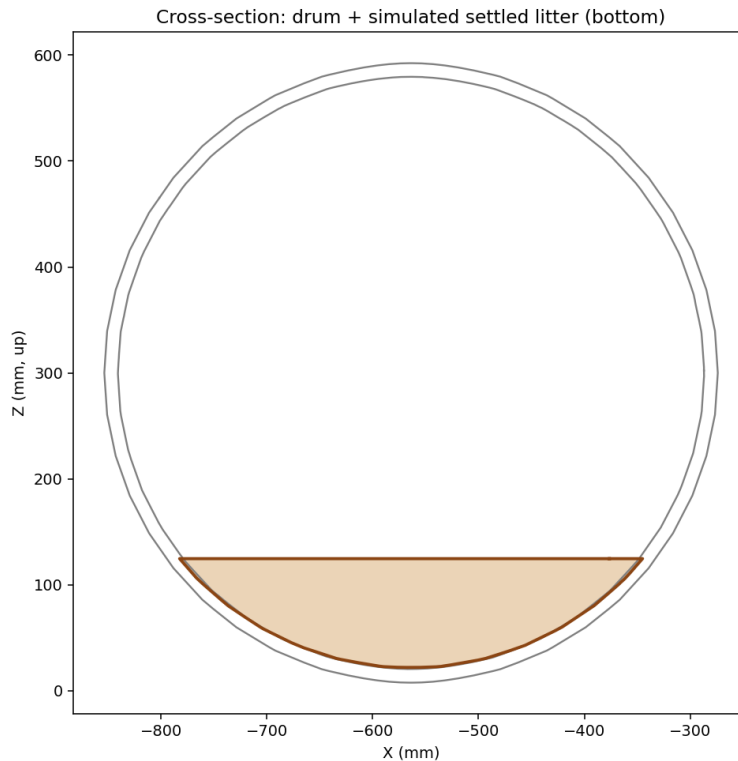
AutoPooper 3000 — Litter Capacity Estimate

Drum volume analysis based on the front-lip drum model and litter-fill simulation mesh

Method

The AutoPooper 3000 drum (*AutoPooper_Drum_FRONT_RAISED.stl*) is a hollow spherical shell. A separate litter-simulation mesh (*AutoPooper_Drum_Estimations.stl*) — modeled as the settled puddle shape litter naturally forms under gravity at the bottom of a rotating drum — was positioned inside the drum: centered on the drum's footprint and seated on the drum's true interior floor (measured directly via ray-casting against the drum geometry, not just its bounding box). The volume of that seated shape is then converted to a weight using published bulk-density figures for standard clumping bentonite clay litter.

Placement Verification



Cross-section through the drum center. Gray: drum shell (inner + outer wall). Tan: simulated settled litter, seated at the bottom of the interior cavity.

Results

Volume	
Liters	8.37 L
US quarts	8.84 qt
US gallons	2.21 gal
Cubic feet	0.296 ft ³

Weight (standard clumping bentonite clay litter)	
Density assumption	kg / lb
Low (810 g/L)	6.78 kg / 14.94 lb
Mid (845 g/L) — best estimate	7.07 kg / 15.59 lb
High (880 g/L)	7.36 kg / 16.23 lb

Notes

- This is a fill-depth estimate for standard clumping bentonite clay litter — the most common type used in automatic self-cleaning litter boxes. Other litter types (silica crystal, pine/wood pellet, paper, tofu/corn) have significantly different densities, roughly 400–650 g/L, and would weigh 3.3–5.4 kg (7.4–12.0 lb) for the same volume.
- 8.4 quarts / ~15.6 lb sits within the typical manufacturer-recommended fill range (roughly 6–10 lb) for automatic globe-style litter boxes of comparable size, which is a reasonable sanity check on the drum's simulated fill volume.
- The litter-simulation mesh represents an idealized flat-settled puddle; real litter will mound and shift with rotation, so actual usable capacity may vary somewhat from this figure.

Sources

- Bulk Density of Cat Litter: 810–880 g/L — Unscented Natural Bentonite Technical Data Sheet ([scribd.com/document/898142116](https://www.scribd.com/document/898142116))
- Cat Litter Bulk Density: Bentonite, Tofu & Cassava g/cm³ Guide — jinansilkpet.com/blog/cat-litter-bulk-density